EECE 576N, From Markov Decision Professes to Deep Reinforcement Learning (3 Credits)

2026, September - December

Mondays & Wednesdays, 2:00 pm – 3:30 pm PST

EOS Room 135 ~~SWNG Room 108~~

Land Acknowledgement

UBC's Point Grey Campus is located on the ancestral and unceded territory of the xwməθkwəy̓əm (Musqueam) people. The land it is situated on has always been a place of learning for the Musqueam people, who for millennia have passed on their culture, history, and traditions from one generation to the next on this site.

Course Information

Course Code, Section, and Title	Academic Period	Credits	Course Delivery Mode
EECE_V 576N – Electrical Engineering Seminar and Special Problems – From Markov Decision Processes to Deep Reinforcement Learning	2026-27 Winter Term 1 (UBC V)	3	In person. Lectures will not be recorded.

Prerequisites and Corequisites: This course assumes a strong background in linear algebra and calculous, as well as in probability and stochastic processes. Prior exposure to basic machine learning or optimization is helpful but not required. No previous experience with MDPs or reinforcement learning is assumed. Familiarity with Python programming is expected for assignments and projects.

Course Description

This graduate-level course provides a comprehensive introduction to sequential decision-making, from foundational theory and algorithms for Markov decision processes (MDPs) to state-of-the-art methods and applications of deep reinforcement learning (RL). Students will learn to model decision problems as MDPs and partially observable MDPs, understand core solution methods (dynamic programming, Bellman equations, optimal policies), and implement practical algorithms (value iteration, policy iteration, linear programming solutions, and analytic solutions in special cases). The latter half of the course transitions to

reinforcement learning: starting with multi-armed bandits, temporal-difference learning, and several landmark algorithms in RL, before moving into deep reinforcement learning. The final portion of the course will expose students to relevant state-of-the-art results and research directions. This course emphasizes mastery of foundational concepts and algorithms, while also exposing students to a broad range of advanced methods at a survey level. Students will gain depth through the project, which allows them to explore a specific method or research direction in more detail. Throughout the course, we will emphasize connections between control theory, fundamental RL algorithms, and modern deep learning implementations. We will also connect the theory and algorithms studied throughout the course to applications in robotics and autonomous decision-making through a course project. By the end of the course, students should be able to formulate real-world problems as MDPs, apply and implement both classical and deep RL algorithms, and critically analyze current research.

Instructor

Cyrus Neary

Contact Details	Office	Office Hours
Please ask all course related questions (including logistic questions) in class, in office hours, or on Piazza (the Piazza link is in Canvas and in the "course materials" section below). For confidential issues, or urgent concerns, please contact me directly by email: cyrus.neary@ubc.ca .	Kaiser 3048	After class on Wednesdays, from 3:30 PM – 4:30 PM in Kaiser 3048.

Instructor's Bio

Bio: Dr. Cyrus Neary is an assistant professor in the Department of Electrical and Computer Engineering at UBC in Vancouver. He earned a BASc in Engineering Physics from UBC, as well as a PhD and MSc in Computational Science, Engineering, and Mathematics from The University of Texas at Austin. Prior to re-joining UBC as a professor, he worked as a postdoctoral researcher at Mila – The Québec AI Institute, in Montréal. His research develops scalable, verifiable, and compositional AI systems for robotics, intelligent autonomy, and computational engineering.

Teaching Assistant

- Sayem Nazmuz Zaman

Course Learning Outcomes/Objectives

Upon the successful completion of this course, students will be able to:

1. Model sequential decision-making problems.
2. Understand and implement fundamental solution concepts and algorithms for MDPs.
3. Understand and implement fundamental reinforcement learning concepts and algorithms.
4. Connect theoretical insights to practical implementations and experiments.
5. Be familiar with landmark deep reinforcement learning algorithms and techniques.
6. Be familiar with advanced topics from current research in the area and be able to critically assess current research.
7. Connect theory and algorithms with hands-on applications.

Learning Materials

The course content will draw on a variety of learning materials including textbooks, research papers, and online code repositories. The two textbooks we will be primarily using at the start of the course are:

- Puterman ML. *Markov decision processes: discrete stochastic dynamic programming*. John Wiley & Sons; 2014.
- Sutton R.S. and Barto A.G. *Reinforcement learning: An introduction (2nd ed.)* The MIT Press; 2018.

Specific research papers, code repositories, and other resources will be shared with the students as they become relevant to the course topics throughout the semester.

All lecture slides will be shared with the students through Canvas.

Piazza: This term we will be using Piazza for class discussion. The system is designed to help you get fast and efficient feedback from classmates, the TA, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza. If you have any problems or feedback for the developers, email team@piazza.com.

Find our class signup link at: https://piazza.com/ubc.ca/winterterm12026/eece576n

Assessment Tools: In-class attendance, polls, and quizzes will use IClicker Cloud platform. Please set up an IClicker student account to participate in these class activities.

- https://lthub.ubc.ca/guides/iclicker-cloud-student-guide/
- https://join.iclicker.com/JOZZ

Course Structure and Learning Activities

This is an in-person course. Learning activities will primarily consist of lectures and in-class discussions. Students will apply course content through homework assignments, and a course project. The course will include in-class quizzes and a midterm exam. The course project will be completed in groups of three students, and aims to prepare students to complete related independent research projects.

Course Schedule

The following is a schedule of topics and deadlines, organized by week. This schedule is subject to change. However, any changes to the schedule will be clearly communicated to the students during lecture.

- Week 1 (September 9th): Course introduction. Introduction to Markov decision processes.
- Week 2 (September 14th, 16th): MDPs continued. Policy classes, value functions, and examples. Intro to Bellman's equations, optimality criteria, value iteration and policy iteration.
    - Introduction to the course project.
- Week 3 (September 21st, 23rd): Linear programming solutions to MDPs. Connections to optimal control. Modeling continued, partially observable MDPs.
- Week 4 (September 28th, 30th): The reinforcement learning problem setting. Balancing exploration and exploitation in bandit problems.
    - Project proposals due (October 2nd).
- Week 5 (October 5th, 7th): Estimating value via Monte-Carlo and temporal difference methods.
- Week 6 (~~October 12th~~ (Thanksgiving), October 14th): Tabular RL algorithms. Q-learning and SARSA.

- Week 7 (October 19th, 21st): Introduction to model-based RL.
- Week 8 (October 26th, 28nd): Intro to supervised learning and function approximation. Function approximation in RL. Intro to deep RL.
- Week 9 (November 2nd, 4th): Policy gradient algorithms.
    - Course project updates due (November 6th).
- Week 10 (November 9th, 11th): Midterm break.
- Week 11 (November 16th, 18th): Midterm exam (16th in class). Landmark algorithms in deep reinforcement learning (TRPO, PPO).
- Week 12 (November 23rd, 25th): Landmark algorithms in deep reinforcement learning (DDPG, TD3, SAC).
- Week 13 (November 30th, December 2nd): Modern topics in deep RL. Project presentations and course wrap up.
- Week 14 (December 7th): Project presentations and course wrap up.
    - Project report due (December 11th).

Assessments of Learning

Assessment will be based on a combination of in-class participation and quizzes, weekly assignments, a midterm exam, and a course project. Homework assignments will reinforce core concepts from lectures and readings. They will include a mix of theoretical questions and programming exercises to assess technical understanding and introduce tools that support both the course project and future research. The midterm exam will be held in class, and will assess student's understanding of the course's core concepts. The course project will allow students to explore an application or extension of course concepts in depth and aims to prepare students to complete related independent research projects. The project will be completed in teams of 3 students and include a short proposal (graded and required for approval by the course instructor), a midterm progress update, and a final written report and in-class presentation. There will be no final exam.

- 5% - In class participation and quizzes.
- 20% - Take-home homework and programming assignments.
- 25% - Midterm exam (November 16th in class).
- 50% - Course project.

- 5% - Project proposal (due October 2nd).
- 5% - Progress update report (due November 6th).
- 5% - Participation in weekly project update survey (every Wednesday at the end of lecture), and individual contributions to team (evaluated via survey responses). Sustained, major under-contribution shown consistently in these surveys may also reduce a student's report and presentation grades by up to 15% for that individual, at the instructor's discretion.
- 10% - Final project presentation (December 2nd and 7th in class).
- 25% - Final project report (due December 11th).

Course Policies

- **Attendance:** Regular attendance is expected from all students. In-class discussion and participation are an important part of this course's learning activities. I acknowledge and accept that unavoidable conflicts with the course may occasionally arise. If you must miss a class, please notify me in advance by email. Repeated or unexplained absences will affect your marks on course participation.
- **Late Submission and/or Missed Work:** Any late or missed work will receive a score of zero. There will be no exceptions to this rule. The lowest homework score will be dropped.
- **Classroom environment:** Respect your peers and contribute to a positive learning environment. We all come from different backgrounds and levels of experience. Please be supportive, inclusive, and constructive in your interactions. In-class discussions are highly encouraged, but please approach them with professionalism and respect. Disruptive behavior, dismissive comments, or lack of courtesy and respect to others will not be tolerated.
- **Academic Integrity/Mis-conduct**: All work you submit in this course must represent your own understanding and effort. Collaboration on homework concepts and problem-solving strategies is encouraged, but all written code, reports, proofs, and explanations must be completed individually. Copying solutions from peers or online sources constitutes academic misconduct. Please acknowledge any resources (e.g., papers, code repositories) that informed your work, and write explanations in your own words.
- **Use of Generative AI**: Grading will allow for generative AI models to be used as a research tool. However, all submitted work must be produced by the students and reflective of their own ideas. Submitted assignments or course project materials that are produced primarily by generative AI will receive a score of zero.

Policies and Resources to Support Student Success

University Policies and Resources

UBC provides resources to support student learning and to maintain healthy lifestyles but recognizes that sometimes crises arise and so there are additional resources to access including those for survivors of sexual assault. UBC values respect for the person and ideas of all members of the academic community. Harassment and discrimination are not tolerated nor is suppression of academic freedom. UBC provides appropriate accommodation for students with disabilities and for religious, spiritual and cultural observances. UBC values academic honesty and students are expected to acknowledge the ideas generated by others and to uphold the highest academic standards in all of their actions.

Details of the policies and how to access support are available on **the UBC Senate website**.